

Scalability & Future Plan

Date	15 March 2024
Team ID	UNM293SmartStudyAI
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Dependence on Gemini API rate limits and quotas for story and image generation	Slower generation and possible failures when many users create comics at the same time	High
2	Character and art-style consistency across comic panels is limited	Panels may look inconsistent, reducing the quality of the story	High
3	Limited language support and no collaborative or export options (e.g., PDF/print)	Restricts reach to a wider audience and sharing of finished comics	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single-instance app handling a small number of concurrent users	Deploy on cloud with auto-scaling and load balancing; add request queues to manage Gemini API calls
2	Data Storage	Stories and generated panels stored locally or in session	Move to cloud database and object storage for user stories, images and history, with caching
3	Performance	Panels generated sequentially, causing wait times	Asynchronous and parallel panel generation, caching of repeated prompts, and image compression
4	Security	Basic API key handling and no user accounts	Secure key management, user authentication, HTTPS, rate limiting and content safety filtering

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Cloud deployment, user accounts and saved comic library	Month 1-2	Reliable access and persistent user data

Phase 2	Character consistency tools and multiple art styles	Month 3-4	More professional, consistent comics
Phase 3	Multi-language support and PDF / print export	Month 5-6	Wider audience and easy sharing
Phase 4	Collaborative story editing, community gallery and mobile app	Month 7-9	Higher engagement and user growth